

Part B : Design Normalized Database

For the relation represented by all of the columns in the CSV file, define all functional dependencies and list them.

Original Functional Dependency

(iid → date, airline, flightnumber, dep_airport, incidentType, severity, delay, num_injuries, reported_by, and aircraft)

iid is the only unique identifier (PK) for each incident therefore it make sense that the iid can be use to determine all other attributes associated with that incident.

Potential Functional Dependency

(airline, and flight number → aircraft, dep_airport)

A flight operated by a specific airline with a specific flight number determines which airpoert it depart from and the type of aircraft it is used.

(airlineid → airline name)

Consider adding airline id - each airline name should could be determine with additional airline id. For example airline id "a1" is assign to maybe AA which represent American Airline. We can create airline table and this will be the functional dependency.

(airportid → airport code)

Similar situation where consider adding airport id - so each airport id can be uniquely determine an airport code. For example "ap1" can determine airport code "JFK" which directly refer to that specific airport. We can create a airport table and that will be the functional dependency.

(Flightid → flight_number, aircraft_model, airlineid (FK), airportid(FK))

Adding Flightid can help determine each flight's operational detail. An example could be a flightid "f1" can determine which airline operate it, the flight number it use, aircafr model, and the departure airport. We can create a flights table and this could be the functional dependency.

(iid → date, incident_type, severity, delay minute, num_injuries, reported by, flightid (FK))

The incidentid (iid) determiness all unique incident detail including which flight it is involved. In this case "i1001" can determine when incident happen, which flight is involve in by flightid (since airportid is FK it also have airport information), the type of incident, severity level, delay time, number of injuries, and who report it. This functional dependency will remain in the incident table.

Transitive Dependency

iid → flightid → airlineid → airlinename

iid → flightid → airportid → dep_airport

iid → flightid → flight_number → aircraft_model

Using the functional dependencies and the rules of normalization, decompose the relational from the CSV into several relations that all satisfy 3NF; give the relations reasonable names.

Explanation

Incident Entity:

(iId (PK), incidentDate, incidentType, severity, delayMinute, numInjuries, reportedBy, flightId (FK))

The Incident entity contain atomic value with no repeating group which satisfy 1NF. The iid is the only PK in the table therefore no partial dependencies will be there. It satisfy 3NF because all non key attributes are depend directly on iid PK without interdependencies and therefore there are no transitive dependencies in the entity.

Airline Entity:

(airlineId (PK), airlineName)

The airline entity is in 1NF because all attributes contain atomic values with no repeating group. It satisfy 2NF by having a single attribute PK airlineid. Lastly it has no transitive dependencies since airline attributes is the only one and it depends directly on airlineid. There are no non prime attributes that can determine other therefore it is in 3NF.

Airport Entity:

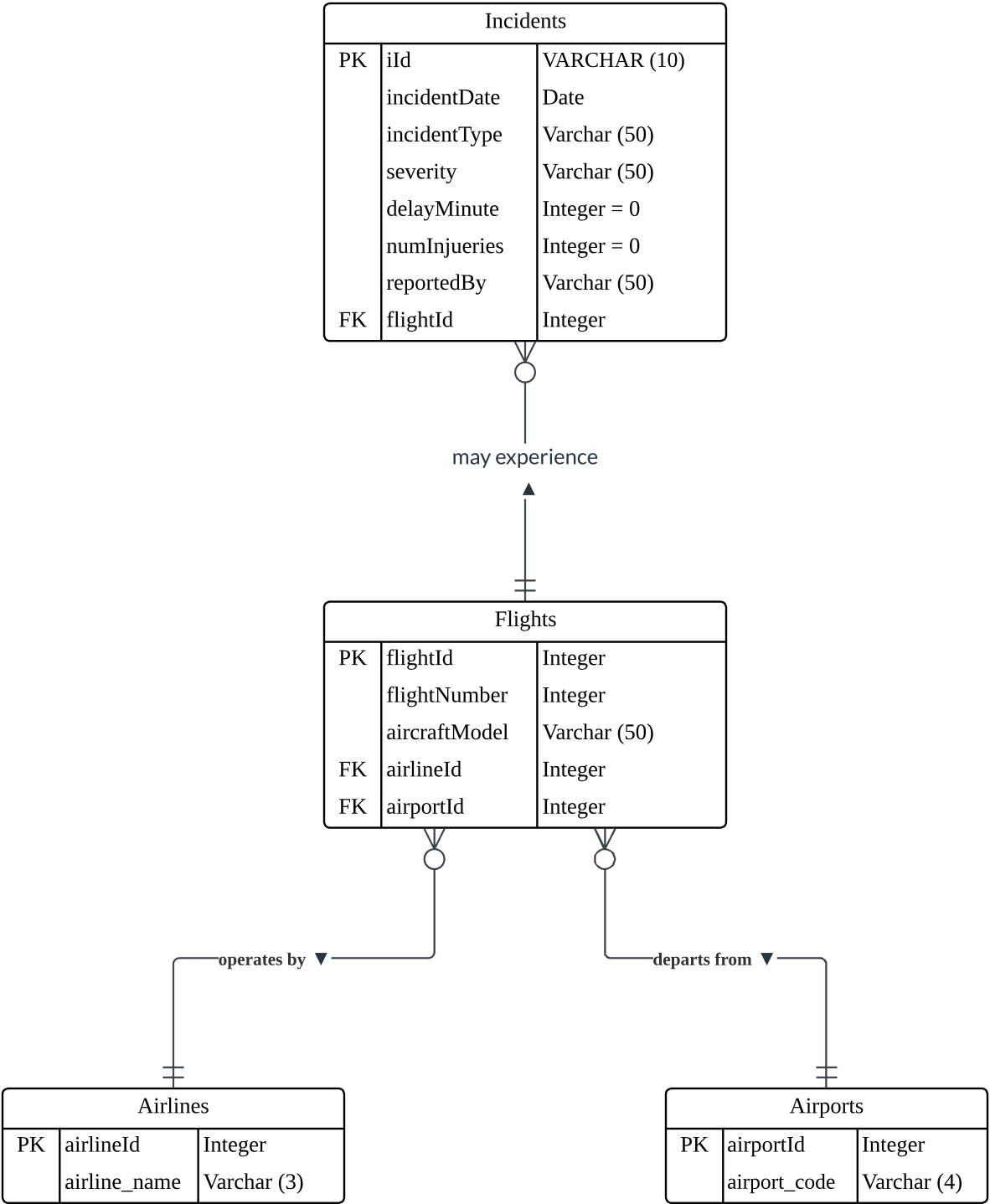
(airportId (PK), airportCode)

Similar to airline entity, airport entity is already in 1NF because all attributes contain atomic values with no repeating group and satisfy 2NF by having single attribute PK airportid. It is in 3NF because there are no transitive dependencies as airport is the only attributes that depend directly on the PK airportid. No attribute determine other attributes and no transitive dependencies is observed.

Flight Entity:

(flightId (PK), flightNumber, aircraftModel, airlineId (FK), airportId (FK))

The flight entity satisfy has contain atomic value across all attributes (1NF). Flightid is the only PK which avoid any partial dependencies. It satisfy 3NF since all non key attribute depends directly on flight id. airportid and airlineid (FK) reference other entity but did not create a dependencies between non key attribute within flight entity. No transitive dependency observe.



For the relations resulting from the normalization, create an ERD in the IE (Crow's Feet) notation. Add all attributes, attribute name, primary and foreign keys, data types, and entity descriptions.

Airlines:

The entity represent airlines that operates flights. It contain airlineid as identification (PK) and include only airline name as non key attributes. It has a one to optional many relationship with flight entity which represent one airline can operates zero or many flights while each flights can only operate by mandatory one airline.

Airports:

The entity represent airports that the flight is departing from. It contains airlineId (PK) as identification and include airportCode as non-key attributes. It has one to optional many relatinshio with flights indicates one airports is a departure point for many flights while each flights can only departs from mandatory one airport

Flights:

The entity represent flight operation. It contain flightId (PK) as identification with non-key attributes flightNumber, aircraftModel, and FK airlineId and airportId. It has a one to optional many relationship with incident table indicates one flight operation may have none or many incident over time.

Incident:

The entity represent bird strikes incident. It contain iId (PK) as identification and include incidentDate, incidentType, severityy, delayMinute, numInjuries, and reportedBy follow with an FK flightId. It has a optional many to mandatory one relationship with flights where indicates even many incidents could occur to one flight but each one of those incidents could only happen to mandatory one flight at each time.

Part B : Design Normalized Database (With Look up Table)

For the relation represented by all of the columns in the CSV file, define all functional dependencies and list them.

Original Functional Dependency

(iid → date, airline, flightnumber, dep_airport, incidentType, severity, delay, num_injuries, reported_by, and aircraft)

iid is the only unique identifier (PK) for each incident therefore it make sense that the iid can be use to determine all other attributes associated with that incident.

Potential Functional Dependency

(airline, and flight number → aircraft, dep_airport)

A flight operated by a specific airline with a specific flight number determines which airpoert it depart from and the type of aircraft it is used.

(airlineid → airline name)

Consider adding airline id - each airline name should could be determine with additional airline id. For example airline id "a1" is assign to maybe AA which represent American Airline. We can create airline table and this will be the functional dependency.

(airportid → airport code)

Similar situation where consider adding airport id - so each airport id can be uniquely determine an airport code. For example "ap1" can determine airport code "JFK" which directly refer to that specific airport. We can create a airport table and that will be the functional dependency.

(Flightid → flight_number, aircraft_model, airlineid (FK), airportid(FK))

Adding Flightid can help determine each flight's operational detail. An example could be a flightid "f1" can determine which airline operate it, the flight number it use, aircrafr model, and the departure airport. We can create a flights table and this could be the functional dependency.

(iid → date, incident_type, severity, delay minute, num_injuries, reported by, flightid (FK))

The incidentid (iid) determiness all unique incident detail including which flight it is involved. In this case "i1001" can determine when incident happen, which flight is involve in by flightid (since airportid is FK it also have airport information), the type of incident, severity level, delay time, number of injuries, and who report it. This functional dependency will remain in the incident table.

Transitive Dependency

iid → flightid → airlineid → airlinename
iid → flightid → airportid → dep_airport
iid → flightid → flight_number → aircraft_model

Categorical Dependency

iid → flightid → aircraft → (aircraft details)
iid → incidentType → (incident type details)
iid → severity → (severity details)
iid → reported.by → (reporter details)

Using the functional dependencies and the rules of normalization, decompose the relational from the CSV into several relations that all satisfy 3NF; give the relations reasonable names.

Explanation

Primary Table

Incident Entity:

(iid (PK), incidentDate, delayMinute, numInjuries, incidentTypeId (FK), severityId (FK), reportedById(FK), flightId (FK))

The Incident entity contain atomic value with no repeating group which satisfy 1NF. The iid is the only PK in the table therefore no partial dependencies will be there. It satisfy 3NF because all non key attributes are depend directly on iid PK without interdependencies and therefore there are no transitive dependencies in the entity.

Airline Entity:

(airlineId (PK), airlineName)

The airline entity is in 1NF because all attributes contain atomic values with no repeating group. It satisfy 2NF by having a single attribute PK airlineid. Lastly it has no transitive dependencies since airline attributes is the only one and it depends directly on airlineid. There are no non prime attributes that can determine other therefore it is in 3NF.

Airport Entity:

(airportId (PK), airportCode)

Similart with airline entity, airport entity is already in 1Nf because all attributes contain atomic values with no repeating group and satisfy 2NF by having single attribute PK airportid. It is in 3NF because there are no transitive dependencies as airport is the only attributes that depend directly on the PK airportid. No attribute determine other attributes and no transitive dependencies is observed.

Flight Entity:

(flightId (PK), flightNumber, aircraftModel, airlineId (FK), airportId (FK))

The flight entity satisfy has contain atomic value accross all attributes (1NF). Flightid is the only PK which avoid any partial dependencies. It satisfy 3NF since all non key attribute depends directly on flight id. airportid and airlineid (FK) reference other entity but did not create a dependencies between non key attribute within flight entity. No transitive dependency observe.

Lookup Table (Categorical Data)

Incident Type:

(incidentTypeId, incidentTypeName)

The Incident Type entity contain atomic value with no repeating group which satisfy 1NF. The incidentTypeId is the only PK in the table therefore no partial dependencies will be there. It satisfy 3NF because all non key attributes are depend directly on incidentTypeId PK without interdependencies and therefore there are no transitive dependencies in the entity.

Severity:

(severityId, severityLevel)

The Severity entity contain atomic value with no repeating group which satisfy 1NF. The severityId is the only PK in the table therefore no partial dependencies will be there. It satisfy 3NF because all non key attributes are depend directly on severityId PK without interdependencies and therefore there are no transitive dependencies in the entity.

Reported By:

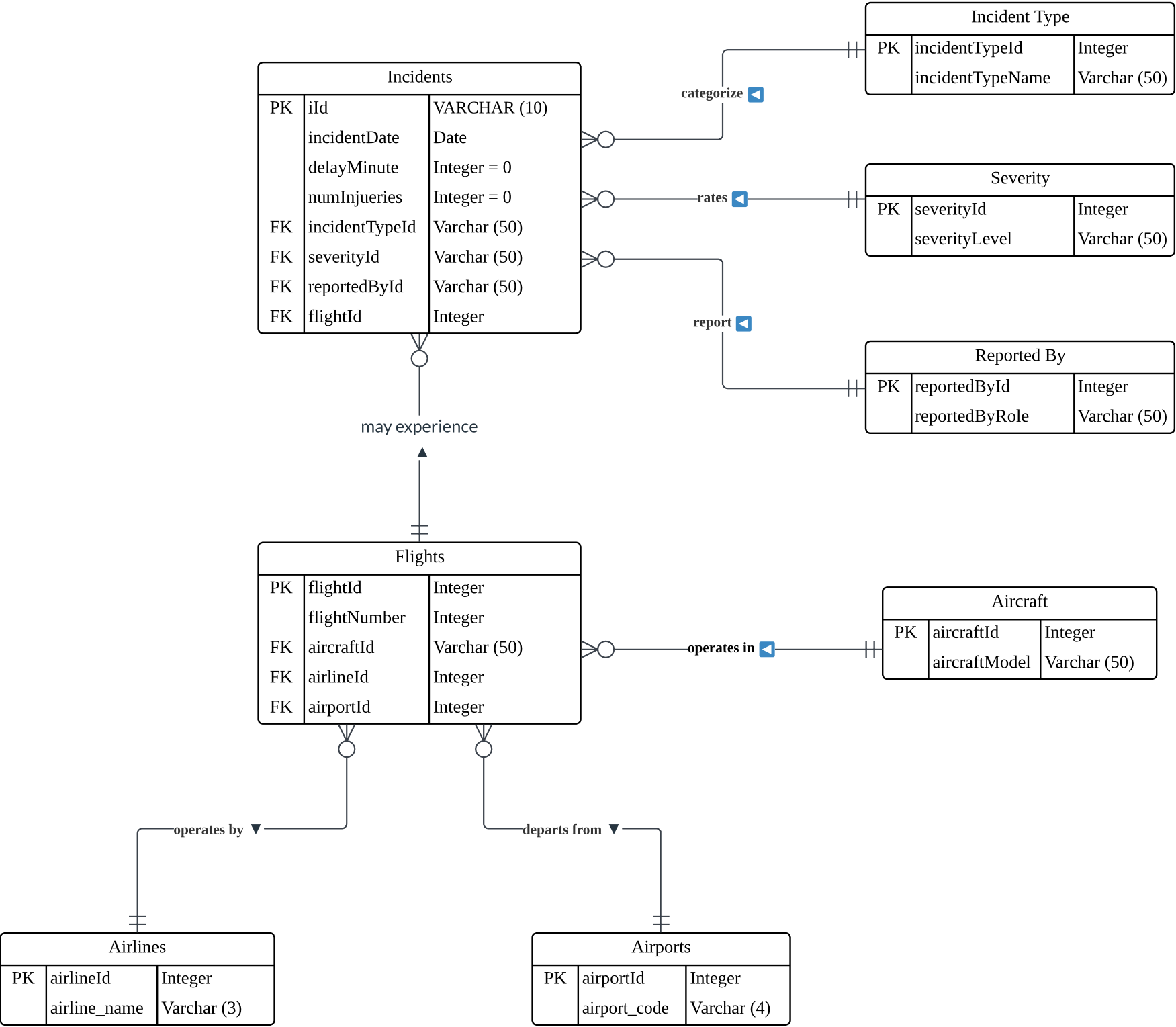
(reportedById, reportedByRole)

The Reported By entity contain atomic value with no repeating group which satisfy 1NF. The reportedById is the only PK in the table therefore no partial dependencies will be there. It satisfy 3NF because all non key attributes are depend directly on reportedById PK without interdependencies and therefore there are no transitive dependencies in the entity.

Aircraft:

(aircraftId, aircraftModel)

The Aircraft entity contain atomic value with no repeating group which satisfy 1NF. The aircraftId is the only PK in the table therefore no partial dependencies will be there. It satisfy 3NF because all non key attributes are depend directly on aircraftId PK without interdependencies and therefore there are no transitive dependencies in the entity.



For the relations resulting from the normalization, create an ERD in the IE (Crow's Feet) notation. Add all attributes, attribute name, primary and foreign keys, data types, and entity descriptions.

Lookup Table

Incident Type:

The entity represent Incident Type lookup table. It contain single PK incidentTypeId and include only incidentTypeName as non key attributes. It has one to optional many relationship with Incident entity represent each incident type can happen on many incident while each incident could only categorize to one incident type.

Severity:

The entity represent severity lookup table. It contain single PK severityId and include only one severityLevel as non key attributes. It has one to optional many relationship with incident entity showing that each severity level can rate on many incident but each incident can only rate to one severity level.

Reported By:

The entity represent Reported By lookup table. It contain single PK reportedById and include only one nonkey attributes reportedByRole. It has mandatory one to optional many relationship with incident entity represent that the person that categorize in one role can report many incidents but each incident can only reported by the person in one role.

Aircraft:

The entity represent aircraft lookup table. It contain single PK aircraftId and have only one non key attribute aircraftModel. It has one to optional many relaitonship indicates that each aircraft model may operates in many flight but each flight can only operate by one plane that categorize in one aircraft model.

Primary Table

Airlines:

The entity represent airlines that operates flights. It contain airlineid as identification (PK) and include only airline name as non key attributes. It has a one to optional many relationship with flight entity which represent one airline can operates zero or many flights while each flights can only operate by mandatory one airline.

Airports:

The entity represent airports that the flight is departing from. It contains airlineId (PK) as identification and include airportCode as non-key attributes. It has one to optional many relatinshio with flights indicates one airports is a departure point for many flights while each flights can only departs from mandatory one airport

Flights:

The entity represent flight operation. It contain flightId (PK) as identification with non-key attributes flightNumber, aircraftModel, and FK airlineId and airportId. It has a one to optional many relationship with incident table indicates one flight operation may have none or many incident over time.

Incident:

The entity represent bird strikes incident. It contain iId (PK) as identification and include incidentDate, incidentType, severityy, delayMinute, numInjuries, and reportedBy follow with an FK flightId. It has a optional many to mandatory one relationship with flights where indicates even many incidents could occur to one flight but each one of those incidents could only happen to mandatory one flight at each time.